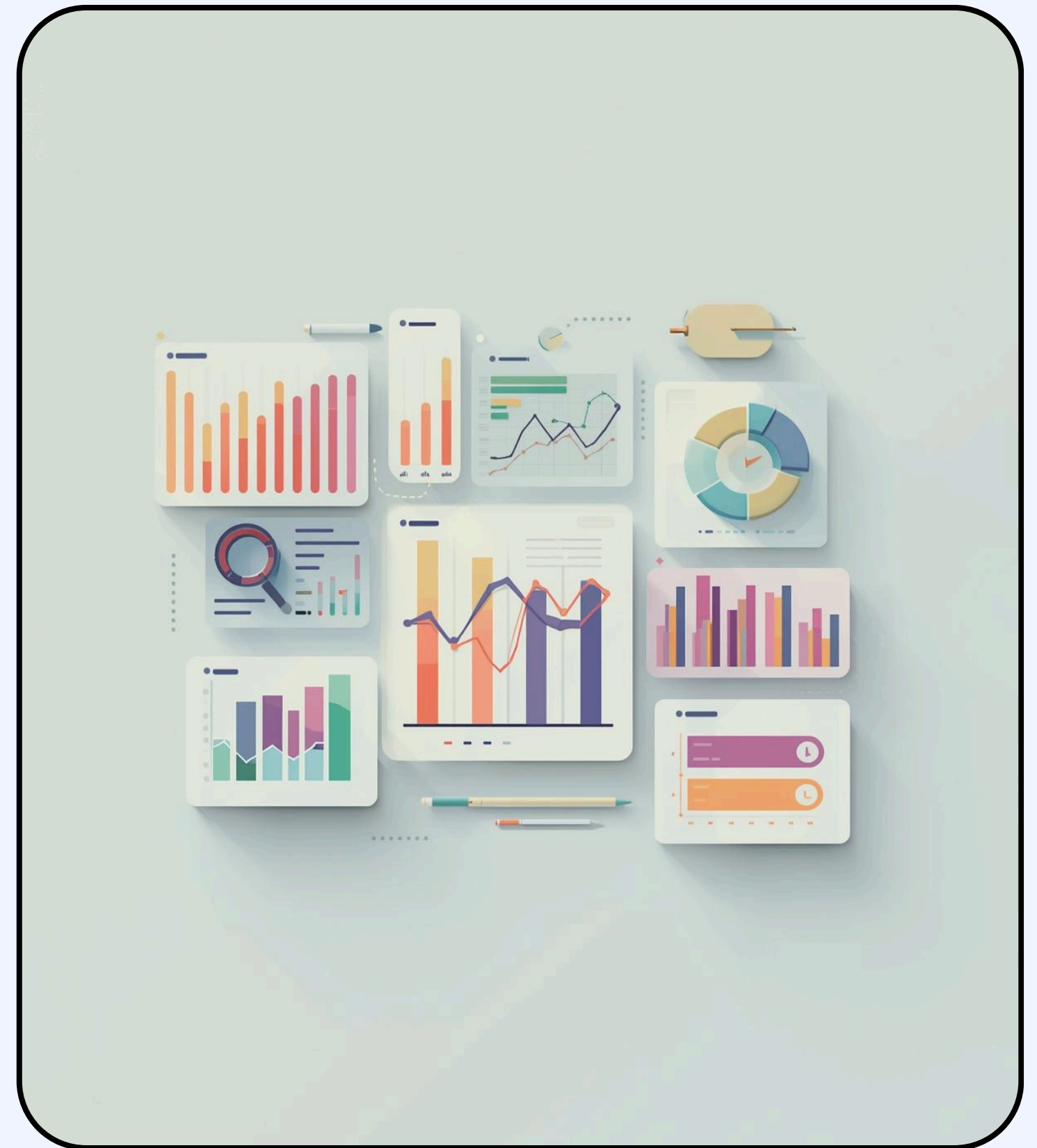
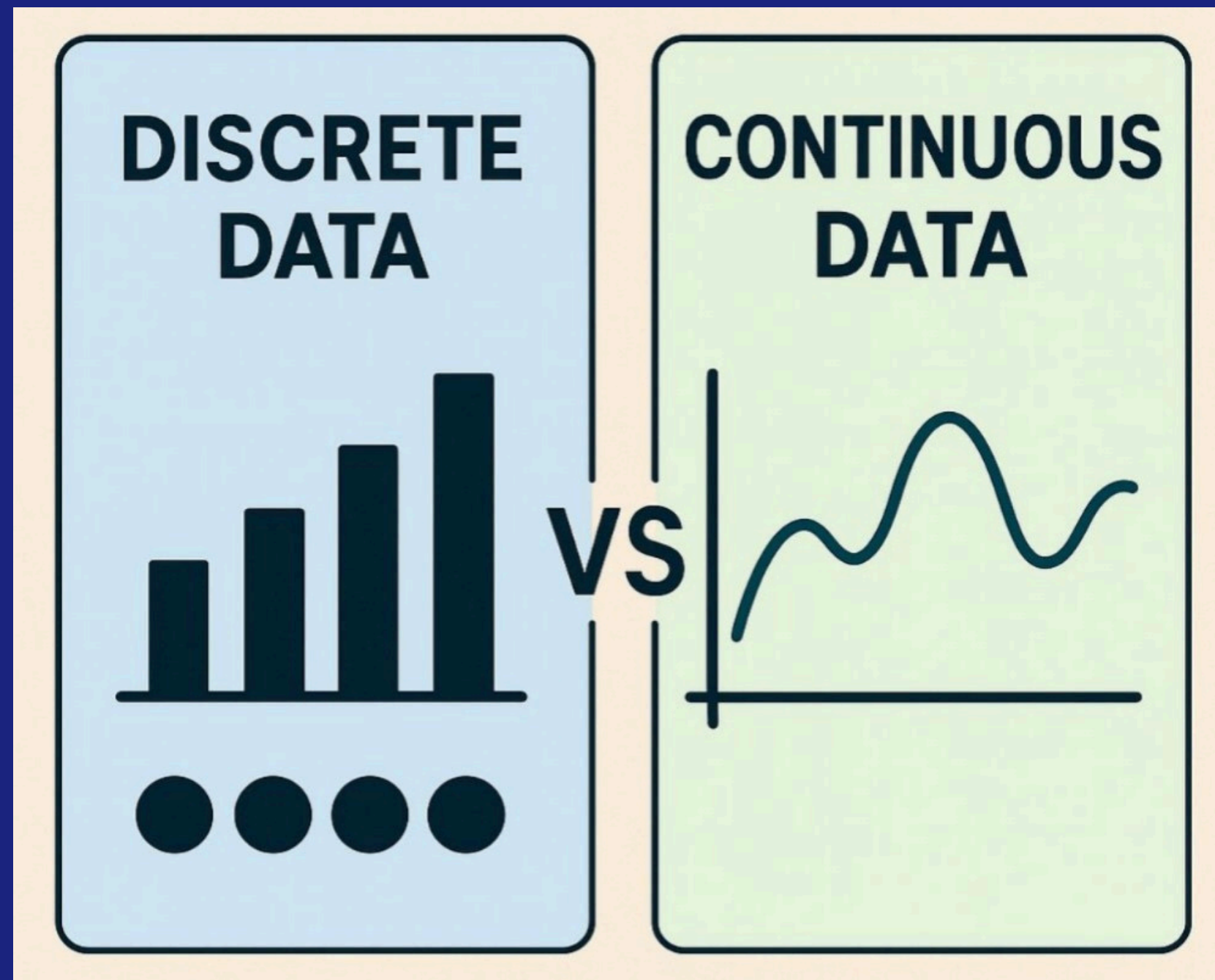


AN INTRODUCTION TO

Data Visualization

Presented by [Shahd Mohammed]





Understanding Data Visualization: Definition and Purpose

Data visualization is the process of converting complex data into **easy-to-understand charts and visuals**, simplifying decision-making.

Why Data Visualization Matters

Fast Understanding

Data visualization allows information to be absorbed quickly, enabling viewers to grasp key insights without sifting through extensive text or data tables.

Pattern Discovery

Visual representations of data reveal trends and relationships that might go unnoticed in raw data, helping to identify critical patterns for analysis.

Better Decisions

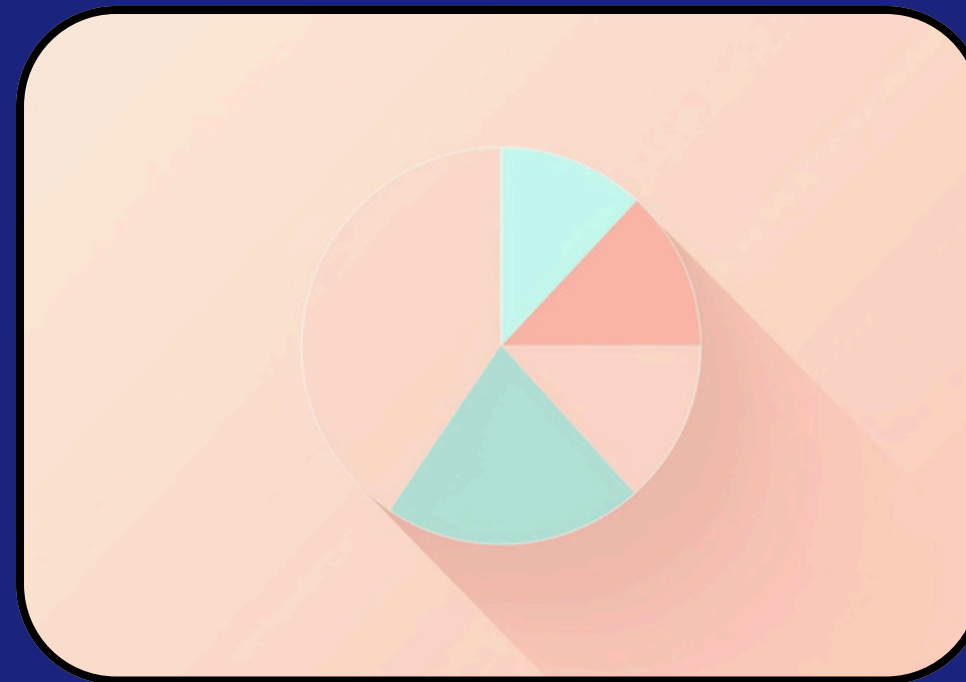
Clear visualizations support smarter decision-making by providing concise insights, allowing stakeholders to understand data implications and make informed choices easily.

Common Visualization Types



Bar Chart

Ideal for comparing categories visually.



Pie Chart

Perfect for displaying percentage distribution easily.



Line Chart

Best for tracking changes across time periods.

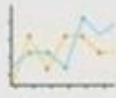
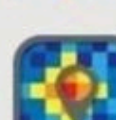
When to Use Each Type and Why:

Selecting the Perfect Chart Type

Accompanying Text :

This comprehensive guide connects chart types with their practical purposes, helping you make informed decisions based on your data needs..

Each chart tells a specific story in a way it excels at.

DATA STORYTELLING			
Chart	Types Of Chart	When To Use It	Example Of Use Case
	Bar Chart	Compare Quantities Across Categories	Compare Sales Of Different Products.
	Line Chart	Show Trends Over Time	Display The Growth Of Website Traffic Over A Year.
	Pie Chart	Highlight Proportion And Percentages	Illustrate The Breakdown Of Expenses In A Budget
	Scatter Plot	Represent Relationships Between Variables	Identify Correlations Between Marketing Spend And ROI.
	Histogram	Visualize The Distribution Of Data	Show The Age Distribution Of Survey Respondents.
	Radar Chart	Compare Multiple Categories Across Dimentions	Evaluate The Performance Of A Product In Various Areas.
	Map	Visualize Geospatial Data	Display Regional Sales Performance On A Map.
	Heatmap	Visualize Data Density And Patterns Especially In Large Data Set	Identifying Hotspots Of Customer Activity In A Shopping Mall.
	Bubble Chart	Represent Three-Dimensional Data	Compare Revenue, Cost, & Profit In Three Dimensions
	Donut Chart	Emphasize Specific Parts Within A Whole	Show The Distribution Of Marketing Expenditures.



Understanding Relationships and Distribution in Data

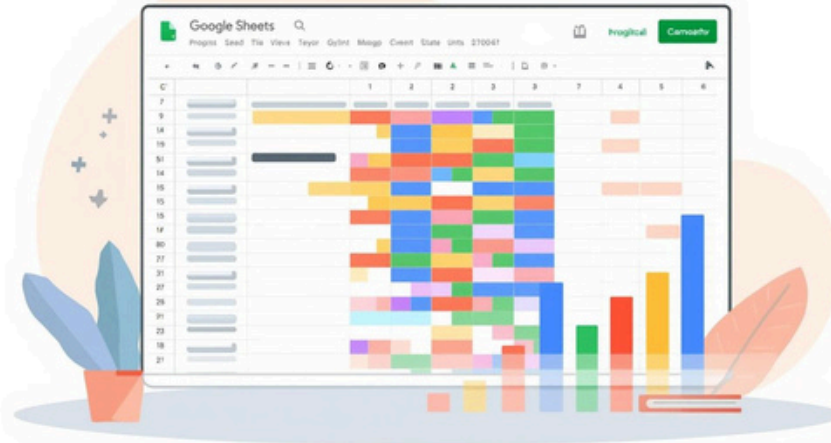
Relationships and distribution visualizations, such as scatter plots and histograms, reveal essential patterns and data trends effectively.

Beginner-Friendly Data Tools



Microsoft Excel

Widely used and easy to learn.



Google Sheets

Free and cloud-based for easy access.

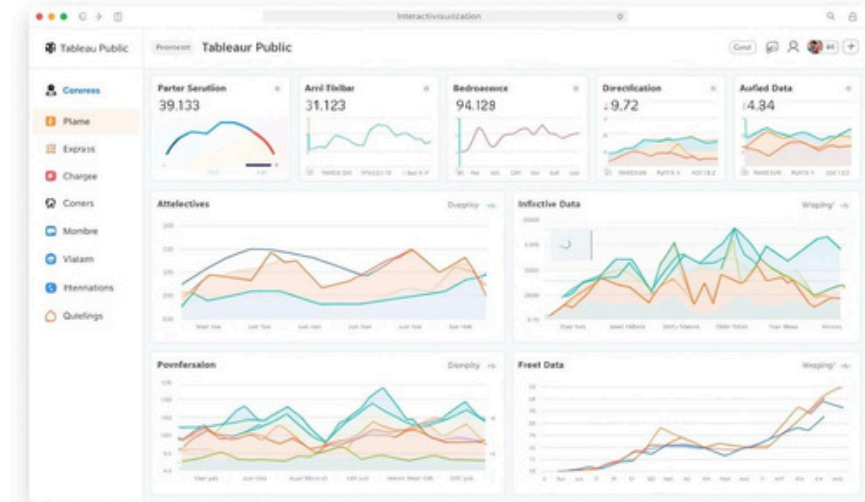


Tableau Public

Powerful visuals, ideal for beginners.



Essential Tools for Data Scientists' Visualization

Data scientists utilize programming tools like Python and R, alongside business tools such as Power BI and Tableau for effective visualization.

How Data Scientists Utilize Visualization



Explore Data

Visuals help identify patterns and anomalies.



Analyze Results

Charts make it easier to interpret outcomes.



Communicate Insights

Clear visuals convey messages to various audiences.



Rules for Effective Data Visualization

Effective visualization requires clarity, simplicity, accuracy, good colors, and a focus on the audience's needs.

Real-World Applications of Data Visualization



Business

Visuals enhance sales and revenue analysis efforts.



Healthcare

Data visualizations track patient outcomes and health trends.



Education

Visual analytics improve student performance assessments.



How to Start Learning Data Visualization

Begin by practicing with personal datasets, learning Excel, and studying effective visualization examples to enhance your skills.



Thank You!

We appreciate your attention and engagement today.

EMAIL

shahdelhewehi@gmail.com

PHONE

0000

SOCIAL MEDIA

@Star Union